SHISHIR POUDEL

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Skills

Languages: C#, Rust, C++, Python, SQL, KQL, Javascript
Frameworks: .NET 8, Django, Node.js, Flutter, SignalR, SvelteKit
Tools: Docker, Git, PostgreSQL, Firebase, Azure, AWS, Linux, CI/CD
Practices: REST APIs, Clean Architecture, Event-Driven Architecture, CQRS, Microservices, Agile

Experience

AI Security Developer | *Cyberclinics, DSU* — Aug 2024 – Present

- Developed and deployed modular Python agents, scanning IPs, URLs, and files against VirusTotal and auto-posting results to Microsoft Defender tickets, resulting in a 40% reduction in false positive investigation
- Built automated playbooks for email phishing analysis, parsing headers and bodies and validating attachments, standardizing Tier-1 alert response workflow and achieving a 55% reduction in response time for phishing incidents
- Fine-tuned small language models (Phi-3) to translate natural language into executable KQL, enabling analysts to query complex log streams and saving approximately 8 hours weekly
- Managed student threat detection team on developing custom yara rules, identifying malicious patterns in network traffic, and integrating with SIEM for real-time alerting

Software Developer | (Contract) *SD Elementary Schools Partnership* — Oct 2025 – Apr 2026

- Led a 6-person Agile team to deploy a web app that automates scoring for literacy assessments, replacing static spreadsheets with a system that tracks historical progress and groups students for immediate academic support
- Structured a data model using Django ORM to track student history, automatically grouping students by skill level to help teachers formulate targeted intervention strategies daily
- Coded React forms that auto-generate based on database configuration, allowing teachers to create new assessment formats without requiring developer intervention

Computer Science Researcher | *Dakota State University* — Aug 2023 – Feb 2024

- Programmed a simulation environment to benchmark 10+ consensus protocols, rigorously quantifying transaction throughput and network fault tolerance to evaluate viability for high-stakes electronic voting

Projects

Enterprise Kanban Board | *.NET 8, SignalR, C#* — Aug 2025 – Dec 2025

- Created a real-time Kanban board application using .NET 8 and Clean Architecture principles, achieving 100% business logic isolation to facilitate comprehensive unit and integration testing
- Enabled real-time synchronization using SignalR to resolve race conditions during concurrent drag-and-drop updates, maintaining sub-42 ms latency for 300 simulated users
- Implemented the Request-Endpoint-Response (REPR) pattern to structure modular API workflows, delivering a functional Trello-style UI with instant activity logging

LLM-powered conversational avatar | *LLaMA 2, Python, Docker* — Aug 2024 – Dec 2024

- Collaborated within a 5-person Agile team to build an interactive virtual avatar that uses LLM-generated speech to drive accurate lip-sync and expressive facial movements, completing 28 user stories
- Reduced latency by 82% under 1000 ms by assembling a concurrent inference pipeline, merging LLaMA2, Wav2Lip, and Text-To-Speech in a docker environment
- Optimized system performance and Wav2Lip inference by implementing concurrent processing techniques (intelligent audio buffer), maximizing throughput to achieve 30 frames per second (FPS) processing speed

Android Mobile App - A Level Notes | *Flutter, Firebase, Google Play* — Feb 2022 – Jun 2023

- Launched a Flutter app on the Play Store that syncs notes and past papers via Firebase, serving over 10,000 users
- Designed a responsive UI with instant search capabilities, allowing students to find Math and Physics topics immediately and contributing to a 4.8-star average rating

Education

BS in Computer Science - Artificial Intelligence (3.84/4.00) | Dakota State University — Dec 2026

Relevant Coursework: Advanced Data Structures & Algorithms, Database Management, Mathematical Foundations of AI, Artificial Intelligence, Software Engineering, Parallel Computing, Programming languages

Awards and Leadership: 1st Place, FBLA Nationals - CS category (Jun 2025) | Outstanding Undergraduate Research Award (Mar 2024) | Math Olympiad Finalist, Nepal (Feb 2022), President - International Student Club DSU (Aug 2024)